Software Requirements Specification

Project: Online Library Management System
Version: 1.0
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1. Introduction

1.1 Purpose

This SRS describes the requirements for an Online Library Management System that allows members to search, reserve, borrow, and return books, while librarians manage inventory, loans, and member records. It is intended for developers, testers, project managers, and library stakeholders.swehb.nasa+1

1.2 Scope

The system will provide web-based access for members and administrators. It will include book catalog management, user registration, search, borrowing, returns, fines, notifications, and basic reporting. Payment gateway integration and procurement management are out of scope for Version 1.0.swehb.nasa

1.3 Definitions

- **Member:** A registered library user.
- **Librarian/Admin:** A staff member who manages books, loans, and users.
- **Loan:** A record linking a book copy to a member for a defined period.
- **Fine:** A charge applied when a book is returned late.

1.4 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.

- Project business requirements document.
- Library operating policies.cse.msu

2. Overall Description

2.1 Product Perspective

The system is a standalone web application with a relational database. It may later integrate with an email/SMS service and institutional authentication system.

2.2 Product Functions

- User registration and login.
- Book catalog browsing and search.
- Book reservation and borrowing.
- Return processing and fine calculation.
- Admin inventory and member management.
- Notifications for due dates and reservations.

2.3 User Classes

User	Main responsibilities
Member	Search books, reserve/borrow items, view loan history
Librarian	Add/update books, issue/return books, manage fines
Administrator	Manage users, roles, reports, and system settings

2.4 Operating Environment

- Web browser: Chrome, Edge, Firefox, Safari.
- Server: Linux or Windows with a supported application server.
- Database: MySQL, PostgreSQL, or equivalent relational database.

2.5 Constraints

- The system must support at least 500 concurrent users.
- Member data must be protected according to applicable privacy policies.
- The interface must be usable on desktop and mobile browsers.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register with name, email, password, and contact number.
FR-02	The system shall authenticate users using email and password.
FR-03	The system shall allow members to search books by title, author, ISBN, subject, or keyword.
FR-04	The system shall display book availability status: available, reserved, borrowed, or unavailable.
FR-05	The system shall allow members to reserve an available book.
FR-06	The system shall allow librarians to issue a book to an eligible member.
FR-07	The system shall set a default loan period of 14 days.
FR-08	The system shall allow librarians to record a book return.
FR-09	The system shall calculate a fine of ₹5 per day for overdue books.
FR-10	The system shall send an email notification three days before a book's due date.
FR-11	The system shall allow administrators to add, edit, and remove book records.
FR-12	The system shall allow administrators to generate reports on loans, overdue books, and active members.

3.2 External Interface Requirements

- **User interface:** Responsive web interface with member and admin dashboards.
- **Hardware interface:** Standard server, barcode scanner optional for book issue/return.

- **Software interface:** Email service for notifications; future integration with institutional login.
- **Communication interface:** HTTPS for all web traffic.

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Search results shall load within 3 seconds for up to 500 concurrent users.
NFR-02	Security	Passwords shall be stored using a secure hashing algorithm.
NFR-03	Security	The system shall enforce role-based access control.
NFR-04	Reliability	The system shall have at least 99% uptime during library operating hours.
NFR-05	Usability	A new member shall be able to search and reserve a book within five minutes without training.
NFR-06	Maintainability	The application shall use modular components and documented code.
NFR-07	Scalability	The system shall support growth from 5,000 to 50,000 book records without redesign.

3.4 Design Constraints

- Use a three-tier architecture: presentation, application, and database.
- Follow RESTful API conventions if a separate frontend and backend are used.
- Use secure HTTPS communication.

3.5 Assumptions and Dependencies

- Users have valid email addresses.
- The library provides accurate initial book inventory data.
- Email delivery depends on the configured third-party email service.

4. Acceptance Criteria

- A member can register, log in, search for a book, and reserve it.

- A librarian can issue and return a book, and overdue fines are calculated correctly.
- Administrators can add books and generate loan reports.
- All critical actions are recorded with timestamps for audit purposes.

5. Appendices

- Use-case diagrams.
- Database schema.
- Wireframes for member search, book details, and librarian dashboard.
- Glossary of library terms.